

Airborne Admin OS — Master Gap Matrix (Phase 6 Audit)

Full remaining-functionality audit • C:\Users\pc\Desktop\Airbone\admin • Read-only (no code modified) • Next.js 15 App Router, Prisma/PostgreSQL, NextAuth, TanStack Query, Zod, Inngest

Method: end-to-end tracing of all 69 Prisma models, ~190 API route files, 22 services, 17 repositories, 20 validation schemas, all dashboard/faculty/portal pages, Inngest functions, plus live verification from Phases 1–5.

Legend

STATUS	Meaning	PRIORITY	Meaning
WORKING	Full trace UI→API→validation→RBAC→service→repo→Prisma→DB works with real data	P0	Production blocker (broken core feature / security)
PARTIAL	Real end-to-end but a meaningful piece is missing	P1	Core functionality incomplete
BROKEN	Trace fails (missing route/UI, method mismatch, wrong enum, security gap)	P2	Important but non-blocking
PLACEHOLDER	Fake/hardcoded surface or honest "not connected" stub	P3	Polish / future
NOT_CONFIGURED	External provider / engine exists but no credentials or not activated		
DEAD	No consumer (route/service/model/UI unused)		
DEFERRED	Intentionally out of scope		

Master Admin OS Gap Matrix

1. Dashboard

MODULE	ROUTE	CAPABILITY	UI	API	VALIDATION	RBAC	DB	REAL DATA	STATUS	PRIORITY	REQUIRED FIX
Dashboard	/	KPI cards + revenue	(dashboard)/page.tsx:39-67	api/v1/dashboard/stats/route.ts:9-57	–	guard("read","analytics")	real	yes	WORKING	–	–
Dashboard	/	Recent leads + quick actions	page.tsx:40-48,125-206	/api/v1/leads	y	y	real	yes	WORKING	–	–
Dashboard	/crm/dashboard	Alias	redirect page	–	–	–	–	–	DEAD	P3	Intentional redirect to /; keep documented

2. Leads / CRM

MODULE	ROUTE	CAPABILITY	UI	API	VALIDATION	RBAC	DB	REAL DATA	STATUS	PRIORITY	REQUIRED FIX
Leads	/leads	List + create	leads/page.tsx	api/v1/leads/route.ts GET/POST	y	guard+org+counselor scope	real	yes	WORKING	–	–
Leads	/leads/[id]	Detail, PATCH, soft-delete	leads/[id]/page.tsx	api/v1/leads/[id]/route.ts	y	guardRecord ABAC	real	yes	WORKING	–	–
Leads	/leads/[id]	Timeline activities CRUD	detail page	leads/[id]/activities/**	y	y	real	yes	WORKING	–	–
Leads	/leads/[id]	Assign counselor	detail page	leads/[id]/assign	y	y	real	yes	WORKING	–	–
Leads	/leads/[id]	Convert → Admission	detail page	leads/[id]/convert	y	y	real	yes	WORKING	–	–
Leads	/leads/[id]	Follow-up scheduling	detail page	leads/[id]/follow-up	y	y	real	yes	WORKING	–	–
Leads	list	Scoring + dedup	badge	lead.service.ts:369-418,29-39	y	y	real	yes	WORKING	–	–
CRM	/crm/deals	Pipeline (admission-funnel derived)	crm/deals/page.tsx	api/v1/crm/deals/route.ts:78-83	y	y	real	yes	DEFERRED	P3	Honest deals:false; real deals need new Opportunity model+FK. Relabel UI "Admission Funnel"
CRM	/crm/pipeline	Alias → /crm/deals	redirect	–	–	–	–	–	DEAD	P3	Intentional alias
CRM	/crm/leads	Alias → /leads	redirect	–	–	–	–	–	DEAD	P3	Intentional alias

CRM	/crm/analytics	Totals/monthly/bySource/byStatus/byCounselor	crm/analytics/page.tsx	api/v1/crm/analytics/route.ts:26-229	y	guard+counselor ABAC:32-35	real	yes	WORKING	–	–
CRM	/crm/meetings	Meeting CRUD (leadActivity)	crm/meetings/page.tsx	api/v1/crm/meetings/route.ts	y	guard+counselor scope	real	yes	WORKING	–	–
CRM	/crm/outreach	Template CRUD + delivery logs	crm/outreach/page.tsx:343-432	api/v1/crm/outreach/route.ts	y	guard("notifications")	real	yes	WORKING	–	Logs empty until Inngest active (see \$20)

3. Admissions

MODULE	ROUTE	CAPABILITY	UI	API	VALIDATION	RBAC	DB	REAL DATA	STATUS	PRIORITY	REQUIRED FIX
Admissions	/admissions	Kanban + drag-drop	admissions/page.tsx (dnd-kit)	api/v1/admissions/route.ts GET/POST	y + STAGE_TRANSITIONS	guard+counselor filter	real	yes	WORKING	–	–
Admissions	/admissions	Detail/update	card dialog	api/v1/admissions/[id]/route.ts	y	guardRecord	real	yes	WORKING	–	–
Admissions	/admissions	Stage transitions (+auto Student on ENROLLED, lead CONVERTED)	stage-log dialog	admissions/[id]/stage	y	y	real	yes	WORKING	–	–
Admissions	/admissions	Fee-plan picker + balance recalc	admissions/page.tsx:504-570	admission.service.ts:48-117	y	y	real	yes	WORKING	–	–
Admissions	/admissions	applicationNo generator	–	admission.repository.ts:137-142	–	y	real	yes	WORKING	P3	count+1 non-atomic (unique constraint backs it)

4. Students

MODULE	ROUTE	CAPABILITY	UI	API	VALIDATION	RBAC	DB	REAL DATA	STATUS	PRIORITY	REQUIRED FIX
Students	/students	List + create	students/page.tsx	api/v1/students/route.ts GET/POST	y	guard+org	real	yes	WORKING	–	–
Students	/students/[id]	Detail/update/soft-delete	students/[id]/page.tsx	api/v1/students/[id]/route.ts	y	guardRecord ABAC	real	yes	WORKING	–	–
Students	/students/[id]	Linked admissions	detail page	students/[id]/admissions	y	y	real	yes	WORKING	–	–
Students	create	Event emission	–	student.service.ts:69-83	–	–	real	–	BROKEN	P2	Emits admission/created with student id — should be student/created event

5. Fee Plans / Payments

MODULE	ROUTE	CAPABILITY	UI	API	VALIDATION	RBAC	DB	REAL DATA	STATUS	PRIORITY	REQUIRED FIX
Fee plans	api/v1/fee-plans/**	CRUD	admissions dialogs	fee-plans/route.ts, [id]/route.ts	y (items min 1)	guard("write", "admissions")	real	yes	WORKING	–	No standalone page — admission-centric by design
Payments	api/v1/admissions/[id]/payments	Record payment + receipt no + balance recalc	admissions/page.tsx:358-368	payment.service.ts:21-84	y	y	real	yes	WORKING	–	–
Payments	api/v1/payments/[id]	Update/status	–	payment.service.ts:91-125	y	guard("write", "payments")	real	yes	WORKING	–	–
Payments	Razorpay	Online collection	–	crm/integrations/route.ts:81-86	–	–	–	no	NOT_CONFIGURED	P3	No gateway creds; honest payments:false; webhook settings field unused

6. Documents

MODULE	ROUTE	CAPABILITY	UI	API	VALIDATION	RBAC	DB	REAL DATA	STATUS	PRIORITY	REQUIRED FIX
Documents	api/v1/admissions/[id]/documents	Upload (R2 presign)	admissions dialog	document.service.ts:22-54	y	y	real	yes	WORKING	–	R2 is legacy path (Supabase used for media); keep as-is
Documents	api/v1/documents/[id]	Review approve/reject	admissions dialog	document.service.ts:120-123	y	y (admin re-check)	real	yes	WORKING	–	–
Documents	api/v1/documents/[id]	Rejection reason	–	document.service.ts:125-127	y	y	real	–	BROKEN	P2	Generic Error (→500) instead of ValidationError (→400) when rejection reason missing

7. LMS — Admin Course Builder

Module	Route	Capability	UI	API	Validation	RBAC	DB	Real Data	Status	Priority	Required Fix
LMS	/lms	Course list/create	lms/page.tsx:40,45	api/v1/lms/courses/route.ts	y	guard	real	yes	WORKING	–	–
LMS	/lms/courses/[id]	Tree view + PATCH	lms/courses/[id]/page.tsx:1237-1242	courses/[id]/route.ts	y	y	real	yes	WORKING	–	–
LMS	/lms/courses/[id]	Delete course	–	no DELETE route	–	–	–	–	BROKEN	P1	Add DELETE /courses/[id] + confirm UI (cascade wipes tree)
LMS	/lms/courses/[id]	Stages CRUD+reorder	:700-714	stages/**	y	y	real	yes	WORKING	–	–
LMS	/lms/courses/[id]	Modules CRUD+reorder+duplicate	:720-810	modules/**	y	y	real	yes	WORKING	–	–
LMS	/lms/courses/[id]	Module quiz settings	:739	modules/[moduleId] PATCH	y	y	real	yes	WORKING	–	–
LMS	/lms/courses/[id]	Chapters/Topics CRUD+reorder+move	:761-785	chapters/**, topics/**	y	y	real	yes	WORKING	–	–
LMS	/lms/courses/[id]	Content create	:517	topics/[topicId]/contents	y	y	real	yes	WORKING	P2	url silently defaults to “#” — surface in UI
LMS	/lms/courses/[id]	Questions CRUD + bulk (≤200)	:170-205	questions/**	y	y	real	yes	WORKING	–	–
LMS	/lms/courses/[id]	Bulk curriculum import	:749	courses/[id]/import	y	y	real	yes	WORKING	–	–
LMS	/lms/courses/[id]	Announcements	:1171-1175	announcements/route.ts	y	y	real	yes	WORKING	–	–
LMS	–	Assign teacher to course	–	–	assignCourseTeacherSchema	–	–	–	DEAD	P3	Unused schema; teachers assigned only via batch members

8. LMS — Batches / Timetable / Attendance / Certificates

Module	Route	Capability	UI	API	Validation	RBAC	DB	Real Data	Status	Priority	Required Fix
Batches	/lms/batches	CRUD	lms/batches/page.tsx	batches/**	y	guard	real	yes	WORKING	–	–
Batches	/lms/batches	Members sync (students+teachers, enrollment sync)	batches/page.tsx:100	batches/[id]/members PUT	y	y	real	yes	WORKING	–	–
Timetable	/lms/timetable	CRUD	lms/timetable/page.tsx	timetable/**	y	y	real	yes	WORKING	P3	recurrenceRule column never written (dead field); update lacks endsAt>startsAt recheck
Attendance	/lms/attendance	Mark sessions+records, list	lms/attendance/page.tsx:80-102	attendance/route.ts GET/POST	y	y	real	yes	WORKING	–	–

Attendance	–	Edit/delete session/record	–	no routes	–	y (delete,lms_attendance unused)	–	–	BROKEN	P1	Add PATCH/DELETE for sessions+records
Certificates	/lms/certificates	Issue	lms/certificates/page.tsx:60	certificates/route.ts POST	y	y	real	yes	BROKEN	P0	certNo generated client-side (:28-31 random) → P2002 collision → unhandled 500. Move to server + retry/unique + try-catch
Certificates	/lms/certificates	List	:45	certificates/route.ts GET	–	y	real	yes	WORKING	–	–
Certificates	–	Revoke (REVOKED enum exists) / delete	–	no routes	–	–	–	–	BROKEN	P1	Add revoke/delete API+UI
Enrollments	/lms/courses/[id]	List/enroll	:400-409	enrollments/route.ts	y	y	real	yes	WORKING	–	–
Enrollments	–	Unenroll / accept-reject	–	no route (status only ACTIVE)	–	–	–	–	BROKEN	P1	Add unenroll/drop + lifecycle state
Portal access	/lms	Provision portal login (argon2, links Student↔User)	lms/page.tsx:155	portal-access/route.ts	y	y	real	yes	WORKING	–	–
Admin dash	api/v1/lms/dashboard/[studentId]	Per-student dashboard	no UI	route exists	–	y	real	yes	DEAD	P3	Build page or remove

9. LMS — Faculty + Student Portal

MODULE	ROUTE	CAPABILITY	UI	API	VALIDATION	RBAC	DB	REAL DATA	STATUS	PRIORITY	REQUIRED FIX
Faculty	/faculty	Dashboard	(faculty)/faculty/page.tsx:40	api/v1/lms/faculty/route.ts	–	role gate TEACHER/ADMIN	real	yes	WORKING	–	–
Faculty	/faculty/students	Students list	:27	faculty/students/route.ts	–	y	real	yes	PARTIAL	P2	Raw join rows — student duplicated per batch; no grouping
Faculty	/faculty nav	"Assignments" link	(faculty)/layout.tsx:23	–	–	–	–	–	BROKEN	P1	Mislabeled → links to /lms/timetable; must point to assignments UI
Portal	/portal/*	Dashboard (enrollments, progress, attendance, announcements, bookmarks, attempts)	portal/page.tsx:356	api/v1/lms/me/route.ts	–	y (STUDENT-only layout)	real	yes	WORKING	–	–
Portal	/portal/courses*	Course list + player + progress + bookmarks	portal/courses/**	me/courses/[courseId], me/progress, me/bookmarks	y	y + enrollment check	real	yes	WORKING	–	–
Portal	quiz	Quiz fetch (no answer leak) + submit/grading engine	components/portal/quiz-modal.tsx	me/courses/[courseId]/quiz/[moduleId]	y	y	real	yes	WORKING	–	courseId path param ignored; moduleId governs — verify intentional
Portal	/portal/progress	Progress page	:47	me/route.ts	–	y	real	yes	WORKING	–	–
Portal	/portal/attendance	Monthly/by-subject breakdown	:53-58	me/attendance-breakdown	–	y	real	yes	WORKING	–	–
Portal	/portal/assessments	List + self-report	list only	me/assessments POST	y	STUDENT has write,lms_assessments	real	yes	BROKEN (security)	P0	Student can POST score+status=PASS directly → self-grading unlock vector.

												Gate to quiz-derived sync only or admin/faculty role
Portal	/portal/certificates	List	:22	me/route.ts	–	y	real	yes	WORKING	–	–	
Portal	/portal/certificates/[certNo]/print	Print	:18-31 (server component, direct prisma)	–	–	no ownership/org check	real	yes	BROKEN (security)	P0		Any student can print any ISSUED cert given certNo/verificationCode — add student/org authorization
Portal	/portal/profile	Read-only display	profile/page.tsx:28-46	me/route.ts	–	y	real	yes	PARTIAL	P2		No edit capability
Portal	/portal/assistant	AI tutor	assistant/page.tsx:49	assistant/route.ts	y	y	real	yes	NOT_CONFIGURED	P3		Stub answer without GEMINI_API_KEY (graceful); no usage quota

10. LMS — Quiz State Machine / Assignments / Dead Model Surface

MODULE	ROUTE	CAPABILITY	UI	API	VALIDATION	RBAC	DB	REAL DATA	STATUS	PRIORITY	REQUIRED FIX
Quiz	–	Sequential module unlock (prev PASS)	–	lms.service.ts:876-898	–	–	real	yes	PARTIAL	P2	Cross-stage gating absent; unlockRules JSON stored but never evaluated
Assignments	api/v1/lms/assignments/**	Create/update/submit/grade	no UI at all	assignments/**, submissions/[id]/grade	y	y	real	yes	BROKEN	P1	Faculty cannot view assignments, submissions, or grade — API-complete, UI-absent. No submission-list GET, no assignment DELETE
Assignments	–	Resubmit	–	assignments/[id]/submit	y	y	real	–	PARTIAL	P2	Resubmit resets status but keeps stale score/feedback — clear on resubmit
Inngest	–	LMS events (cert issued, enrollment, attendance SMS)	–	events/functions/**	–	–	–	–	DEAD	P2	Zero LMS handlers in Inngest functions; nothing enqueues LMS events
LmsNotification	–	In-app notifications	–	–	–	lms_notifications perm exists	real (table)	no	DEAD	P2	No service/route/UI — unused table+permission
LmsChatRoom/LmsMessage	–	Chat	–	–	–	lms_chat perm exists	real (table)	no	DEAD	P3	No service/route/UI — deferred feature
Feature gate	–	isLmsEnabled (sms.enabled)	–	src/lib/lms/feature-flag.ts	–	–	–	–	DEAD	P3	Never imported — LMS always accessible

11. LMS — Public Certificate Verification

MODULE	ROUTE	CAPABILITY	UI	API	VALIDATION	RBAC	DB	REAL DATA	STATUS	PRIORITY	REQUIRED FIX
Verify	/verify/[code]	Public cert lookup (certNo OR verificationCode, ISSUED only)	verify/[code]/page.tsx:40	api/v1/public/lms/certificates/route.ts	–	public by design	real	yes	WORKING	–	–

12. CMS — Pages / Sections / Blocks / Versions

MODULE	ROUTE	CAPABILITY	UI	API	VALIDATION	RBAC	DB	REAL DATA	STATUS	PRIORITY	REQUIRED FIX
Pages	/cms	List + create + edit metadata	cms/page.tsx:63-107	pages/**	y	y	real	yes	WORKING	–	–
Pages	/cms	Publish/status transitions	:110-121,298-311,399-414	pages/[id]/publish	y	y	real	yes	WORKING	–	–
Pages	/cms	Version snapshot on publish + list + rollback	no UI	versions/**, [vid]/rollback	y	y	real	yes	BROKEN (API-only)	P1	Server-side complete — build editor UI (sections/blocks/reorder/versions/rollback). cms/page.tsx is list-only
Pages	/cms	Section editor	no UI	pages/[id]/sections/**	y	y	real	yes	BROKEN (API-only)	P1	Same as above
Pages	/cms	Block editor	no UI	sections/[sid]/blocks/**	y	y	real	yes	BROKEN (API-only)	P1	Same + ContentBlock types unseeded → blockType lookup 404s

Pages	/cms	Layout reorder	no UI	pages/[id]/layout	y	y	real	yes	BROKEN (API-only)	P1	Same
Pages	/cms	Live preview	cms/page.tsx:160-171 fake toggle → toast only	–	–	–	–	–	PLACEHOLDER	P2	Replace with real preview (public render/iframe)
Pages	–	Scheduled publish	–	page.service.ts:413-456	y	–	real	yes	NOT_CONFIGURED	P2	Logic exists but driven by cms/scheduled.check Inngest event — never fires (Inngest disabled). Wire cron/inngest
Pages	–	Public propagation	–	api/public/pages	–	public	real	yes	WORKING	–	–

13. CMS — Nav / Content Blocks / Courses / Media

MODULE	ROUTE	CAPABILITY	UI	API	VALIDATION	RBAC	DB	REAL DATA	STATUS	PRIORITY	REQUIRED FIX
Nav	api/v1/nav/**	Menu CRUD + nested items	no UI anywhere	nav/**, nav.service.ts	y	y	real	yes	DEAD (route-only)	P1	Build nav management UI (settings or CMS); add to a public endpoint (currently api/public/settings omits navMenus)
ContentBlock	api/v1/blocks/**	Type registry	no UI	blocks/**, block.service.ts	y	y	real	yes	DEAD (route-only)	P1	Seed block types + registry UI, else block feature unusable end-to-end
Courses	/courses	List + create	courses/page.tsx:55-138	api/v1/courses/**	y	y	real	yes	WORKING	–	–
Courses	/courses	"Go Live" publish	courses/page.tsx:73-80 sends PATCH	courses/[id]/publish POST only	y	y	real	yes	BROKEN	P0	UI method mismatch → 405, button silently fails. Change to POST
Courses	/courses	Edit (SEO only)	:140-154	courses/[id] PATCH	y	y	real	yes	PARTIAL	P1	No UI to edit title/duration/fee/category/eligibility/curriculum/banner after creation
Courses	/courses	Create status select	:314-317	createCourseSchema	stripped	y	real	–	BROKEN	P1	Status not in schema → always DRAFT; remove misleading control or wire it
Courses	/courses	Curriculum editor	:376-401 read-only JSON	–	–	–	real	yes	BROKEN	P1	No add/edit/reorder of modules; build editor
Courses	/courses	Versions + rollback	no UI	versions/**, [vid]/rollback	y	y	real	yes	BROKEN (API-only)	P1	Add versions UI
Courses	/courses	Banner/gallery media picker	missing	course.service.ts:53-117 (MediaUsage)	y	y	real	–	PARTIAL	P2	Schema supports bannerImageId/galleryIds; add MediaPicker
Courses	/courses	Archive delete	:106-115	courses/[id] DELETE	y	y	real	yes	WORKING	P3	Toast says "removed from database" but it archives — fix wording
Courses	–	Scheduled publish	–	course.service.ts:304-350	y	–	real	yes	NOT_CONFIGURED	P2	Same Inngest-driver gap as pages
Courses	–	Public propagation	–	api/public/courses	–	public	real	yes	WORKING	–	–
Media	/media	Upload → Supabase, folders, search, copy, delete (soft+usage guard)	media/page.tsx:70-443	api/v1/media/**	y	y	real	yes	WORKING	–	–
Media	/media	Folder create	no UI	media/folders POST	y	y	real	yes	BROKEN (API-only)	P1	Add "New Folder"
Media	/media	Rename/move asset	no UI	media/[id] PATCH	y	y	real	yes	BROKEN (API-only)	P1	Asset dialog has only copy/delete/open
Media	/media	Replace file	no UI	media/[id]/replace	y	y	real	yes	BROKEN (API-only)	P1	Add replace UI
Media	/media	Bulk delete / multi-select	missing	–	–	–	–	–	BROKEN	P2	Add
Media	/media	Folder asset-count badge	:267-271	media.repository.ts:41-51	–	–	real	–	BROKEN	P3	FOLDER_SELECT lacks assetCount → badge never renders

14. Resources / Blog / Jobs / Testimonials (Phase 5 — live-verified)

MODULE	ROUTE	CAPABILITY	UI	API	VALIDATION	RBAC	DB	REAL DATA	STATUS	PRIORITY	REQUIRED FIX
Resources	/resources	Full CRUD + publish/unpublish/archive + gating + download count	components/cms/resource-manager.tsx	api/v1/resources/**	y	y	real	yes	WORKING	–	–
Resources	/resources	Public + HMAC gated download (20/min/IP)	–	api/public/resources, resource-download	y	token	real	yes	WORKING	–	–
Blog	/blog	Resources manager type=DOCUMENT	blog/page.tsx	api/public/blogs	y	y	real	yes	WORKING	–	Not in sidebar (by design)
Jobs	/jobs, /jobs/[id]	CRUD + publish transitions	jobs/page.tsx, jobs/[id]/page.tsx	api/v1/jobs/**	y	y	real	yes	WORKING	–	–
Job apps	/jobs/[id]	Submit/list/detail	jobs/[id]/page.tsx	api/v1/job-applications/**	y	y	real	yes	WORKING	–	–
Job apps	/jobs/[id]	Status change	:104 offers OFFERED, HIRED	job-applications/[id]/status	zod/DB enum lack OFFERED/HIRED (only SELECTED)	y	real	yes	BROKEN	P0	UI offers statuses the enum rejects → 400. Remove OFFERED/HIRED from UI or add to enum+transition map + badge styles
Testimonials	/testimonials	CRUD + review + featured + avatar + pending badge	testimonials/page.tsx	api/v1/testimonials/**	y	y	real	yes	WORKING	–	P3: no order field UI; no unpublish→PENDING path

15. Users / Roles / Permissions / Auth

MODULE	ROUTE	CAPABILITY	UI	API	VALIDATION	RBAC	DB	REAL DATA	STATUS	PRIORITY	REQUIRED FIX
Users	/users	List/search/create/update/deactivate	users/page.tsx	api/v1/users/**	y	guard + SUPER_ADMIN role-change guard	real	yes	WORKING	–	–
Users	/users	Invite	invite dialog	users/route.ts action=invite → user.service.ts:22-51	y	y	real	yes	PARTIAL	P1	Email never sent (Inngest off + no RESEND_API_KEY); user created isActive:false with unused token; UI toasts success — surface "email not sent"
Users	api/v1/users/invite/accept	Accept invite	no UI	route exists	y	y	real	yes	PARTIAL	P1	No accept-invite page for staff
Roles	/users	Role enum + labels	users/page.tsx:20-42 hardcoded	Prisma UserRole	–	–	real	yes	WORKING	P3	Hardcoded in 2 places; sync risk
Permissions	–	Per-org permission overrides	–	–	–	static matrix only	Permission table UNUSED	no	PARTIAL	P2	Matrix comment claims DB overrides — aspirational; no API/UI to manage Permission rows
Auth	/login	Credentials+orgSlug login, argon2, session revocation	(auth)/login/page.tsx	lib/auth/config.ts:57-112	y	y	real	yes	WORKING	–	–

16. Settings / Organization / Campuses / Profile / Password

MODULE	ROUTE	CAPABILITY	UI	API	VALIDATION	RBAC	DB	REAL DATA	STATUS	PRIORITY	REQUIRED FIX
Settings	/settings	Read org settings	settings/page.tsx:30-33	api/v1/organizations GET	y	y	real	yes	WORKING	–	–
Settings	/settings	Toggles + webhook URLs	:58-101 + "Saved" toast :39	organizations PATCH → org.service.ts:28-56	y	y	real	yes	BROKEN (fake-success)	P0	Values persist into org.settings JSON but NOTHING consumes them: maintenanceMode, applicationIntake, forceDebugLogs, razorpayWebhook, vapiWebhook, whatsappWebhook; no /api/webhooks/* routes; public intake never checks

											applicationIntake. Ops believes controls are enforced when they aren't
Settings	/settings	Env/secrets display	:164-180	–	–	–	–	–	WORKING	–	Display-only by design
Org	api/v1/organizations/**	Tenant scope + SUPER_ADMIN listAll	–	organizations/route.ts:14-20	y	y	real	yes	WORKING	–	–
Campuses	api/v1/organizations/campuses/**	CRUD + code-unique + headCounselor org check + delete guard	–	org.service.ts CampusService	y	y	real	yes	WORKING	–	No dedicated UI beyond settings?
Profile	topbar	"Profile Settings" menu item	topbar.tsx:78-81	users/[id] PATCH self-exempt exists	y	y	real	–	BROKEN (no-op)	P1	No handler/navigation — wire to an edit dialog
Password	–	Change password	no UI	no route (changePasswordSchema at user.schema.ts:31 zero callers)	y (dead)	–	–	–	DEAD	P1	Implement change-password route + UI (self-service)

17. Audit / Activity / Notifications

MODULE	ROUTE	CAPABILITY	UI	API	VALIDATION	RBAC	DB	REAL DATA	STATUS	PRIORITY	REQUIRED FIX
Audit	/audit	Immutable hash-chain writes + list/filter/search + CSV export	audit/page.tsx:33-71	api/v1/audit/route.ts; audit.service.ts	y	guard("read","audit")	real	yes	WORKING	–	–
Audit	–	Coverage	–	written across user/org/campus/lead/media/admission/job/payment/auth services	–	–	real	yes	WORKING	–	LMS mutations largely unlogged
Activity	–	Feed writes	–	activity.service.ts	–	–	real	yes	WORKING	–	–
Activity	–	Feed read/UI	no dedicated page/API	–	–	–	real	–	PARTIAL	P2	No admin surface for activity feed (visible only via CRM detail contexts)
Notify	/crm/outreach	Template CRUD + dispatch engine + logs	crm/outreach/page.tsx	api/v1/crm/outreach/route.ts, notification.service.ts	y	y	real	yes	WORKING	–	Logs only written when Inngest processes (inactive now)
Notify	–	Email (Resend)	–	notification.service.ts:105-151 real fetch	–	–	real	–	NOT_CONFIGURED	P3	No RESEND_API_KEY → all email dead
Notify	–	SMS (Twilio) / WhatsApp (WATI)	–	notification.service.ts:98-103	–	–	–	–	NOT_CONFIGURED	P3	Honest NOT_CONFIGURED; no impl. env.ts marks Twilio "reserved"
Notify	–	In-app inbox	topbar Bell icon no onClick (topbar.tsx:54-57, fake pulsing dot)	no inbox API/route	–	–	–	–	BROKEN (no-op)	P1	Add inbox page + IN_APP delivery + wire Bell handler
Notify	/notifications	Orphan page	notifications/page.tsx:13-24 "Not Connected"	–	–	–	–	–	PLACEHOLDER	P2	Not in sidebar; duplicates real Outreach module — repurpose or remove

18. Workflows / Feature Flags

MODULE	ROUTE	CAPABILITY	UI	API	VALIDATION	RBAC	DB	REAL DATA	STATUS	PRIORITY	REQUIRED FIX
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Workflows	–	Engine	–	–	–	–	Workflow/WorkflowRun tables exist (schema:642,662) — ZERO src references	no	DEAD	P2	No routes/services/UI; admission stages are static enum, not WorkflowRun-driven. Implement or drop tables
Feature Flags	–	Manager	–	–	–	–	FeatureFlag table (schema:1268) unused; flags live as org featureFlags JSON	no	DEAD	P2	No admin UI/API to manage flags; isLmsEnabled reads JSON but is never used

19. Public Forms / Lead Intake

MODULE	ROUTE	CAPABILITY	UI	API	VALIDATION	RBAC	DB	REAL DATA	STATUS	PRIORITY	REQUIRED FIX
Intake	api/public/leads	x-intake-key + 5/min/IP rate-limit + zod + org routing + dedup 409 + leadActivity + UTM + gate token	(public site)	api/public/leads/route.ts:17-175	y	key + rate	real	yes	WORKING	–	–
Intake	–	applicationIntake toggle enforcement	–	api/public/leads/route.ts	–	–	real	–	BROKEN	P0	Toggle never checked — wire org.settings.applicationIntake into intake gate (see §16)
FallbackLead	–	Dead-letter queue	–	–	–	–	table exists (schema:2078) + migration — ZERO src usage	no	DEAD	P3	No queue-fallback path ever writes to it

20. Integrations / External Providers

MODULE	ROUTE	CAPABILITY	UI	API	VALIDATION	RBAC	DB	REAL DATA	STATUS	PRIORITY	REQUIRED FIX
Integrations	/crm/integrations	Status readout (env-derived, honest)	crm/integrations/page.tsx	api/v1/crm/integrations/route.ts:10-103	–	y	env	–	WORKING	P3	Media status computed from R2 env while media uses Supabase — can misreport health
Facebook	facebook/callback	OAuth	page.tsx = plain redirect to /crm/integrations	no token exchange	–	–	–	–	DEAD	P3	No Meta OAuth, no leads pull; flag-only status
Google Ads	–	Ingestion	–	flag-only	–	–	–	–	NOT_CONFIGURED	P3	Public env flag only
Supabase	–	Storage	media upload	storage/supabase.ts real	–	–	real	yes	CONFIGURED	–	Only live external provider (creds in .env)
Inngest	api/inngest	~30 functions served; emitEvent no-op when INNGEST_EVENT_KEY=local	–	events/inngest.ts:4-19, events/functions/**	–	–	–	–	NOT_CONFIGURED	P1	All events silently dropped in current env (invite email, lead events, payment receipts, scheduled publish). Set real key + signing key + dev server
Gemini	/portal/assistant	AI tutor	–	assistant/route.ts	y	y	–	–	NOT_CONFIGURED	P3	Stub without key (graceful)

21. Other Pages

MODULE	ROUTE	CAPABILITY	UI	API	VALIDATION	RBAC	DB	REAL DATA	STATUS	PRIORITY	REQUIRED FIX
Vapi	/vapi	Voice AI logs	vapi/page.tsx:13-24 "Not Connected"	–	–	–	–	–	PLACEHOLDER	P3	Orphan (not in sidebar); settings vapiWebhook unused
Errors	/errors	Error monitoring	errors/page.tsx:13-24 "Not Connected"	–	–	–	–	–	PLACEHOLDER	P3	Orphan; no Sentry/Datadog wiring
Blog	/blog	Resources DOCUMENT	blog/page.tsx	api/v1/resources + api/public/blogs	y	y	real	yes	WORKING	–	–

1. Core Modules Already 100% Functional (A)

1. **Authentication** — credentials+orgSlug, argon2, session revocation, middleware RBAC
2. **Dashboard** — real KPI stats + recent leads
3. **Lead management** — full lifecycle (list/create/detail/timeline/assign/convert/follow-up/scoring/dedup)
4. **CRM analytics / reports** — real Prisma aggregation, counselor ABAC
5. **CRM meetings**
6. **CRM outreach templates + logs** (once Inngest active)
7. **Admissions** — kanban, stage machine, auto-Student/lead sync, fee-plan binding, payments
8. **Students** — CRUD + soft-delete + uniqueness + linked admissions
9. **Fee plans + payment recording** — receipts, balance recalc
10. **Documents** — R2 presign upload + approve/reject review
11. **Placements, Hiring Partners, Jobs** — CRUD, publish, public sync, job applications submit/list/detail
12. **Testimonials, Resources/Blog, Jobs, Course manager (list/create/archive), Media upload/folders/delete** — Phase 5 live-verified
13. **Public APIs** — pages, resources, blogs, courses, jobs, testimonials, placements, settings, leads (rate-limited+keyed), resource-download (HMAC+rate-limited), certificate verification
14. **LMS admin builder** — course tree, stages/modules/chapters/topics/contents, questions+bulk, import, announcements, batches+members sync, timetable, attendance marking, enrollments, portal provisioning
15. **Student portal** — dashboard, course player, progress, bookmarks, quizzes (no answer leak, real grading), attendance breakdown, timetable, announcements, certificates list
16. **Faculty dashboard**
17. **Audit logs** — hash-chained, append-only, filterable + CSV export
18. **RBAC + org isolation** — guard() + guardRecord() ABAC + orgId-scoped repos everywhere
19. **Public lead intake** — keyed, rate-limited, deduped, UTM-aware

2. Core Modules Partially Functional

Module	Missing
Courses (CMS)	Edit is SEO-only; no curriculum editor; no versions UI; create status stripped; banner/gallery picker missing
Page editor	All server-side (sections/blocks/reorder/versions/rollback) — zero UI
Media library	No folder-create/rename/move/replace/bulk UI; asset-count badge dead
Nav menus	API+validation+RBAC complete — zero UI, no public propagation
ContentBlock types	Registry API only, no types seeded, no UI
Users invite	User created but email never sent; no accept-invite UI
Profile/account	Read-only portal profile; admin "Profile Settings" no-op; no password change
Notifications	Real template/log engine but email/SMS/WhatsApp unconfigured; no in-app inbox (Bell fake)
Assignments (LMS)	Fully implemented API — zero UI; faculty link mislabeled
Scheduled publish (pages+courses)	Logic exists, never runs (Inngest disabled)
Module unlock (LMS)	In-module sequential gating works; cross-stage + unlockRules unevaluated

3. Broken Core Features (E)

#	Broken item	Why	Severity
1	Courses "Go Live"	UI sends PATCH to a POST-only route → 405, silent failure	P0
2	Job application status	UI offers OFFERED/HIRED; zod + DB enum only allow SELECTED → 400	P0

3	Settings toggles (intake/maintenance/webhooks)	Persisted but never consumed anywhere; no webhook routes	P0
4	Student self-assessment	STUDENT can POST score+status=PASS via me/assessments → self-grade unlock vector	P0
5	Certificate print page	Direct prisma, no ownership/org check	P0
6	Certificate issue	certNo generated client-side → P2002 collision → unhandled 500	P0
7	Missing delete/revoke routes	Course, assignment, attendance session/record, certificate revoke, unenroll — all absent	P1
8	Assignment UI	API-complete, faculty cannot view/grade	P1
9	Dashboard-student API	No UI consumer	P3
10	Minor issues	document rejection →500; student event wrong name; faculty students duplicate rows; media asset-count badge; timetable update recheck	P2

4. Placeholder / Fake Features

- /cms "Preview" button → toast-only toggle, no preview output
- /notifications, /vapi, /errors — static "Not Connected" orphan pages (not in sidebar)
- Topbar Bell icon — animated pulsing dot, no onClick, no inbox (implies unread that doesn't exist)
- Topbar "Profile Settings" / "System Configurations" — menu items with no handler
- User invite — success toast while email silently drops
- Settings — "Settings Saved" toast for controls that do nothing
- AI tutor assistant — stub answer without GEMINI_API_KEY (declared, honest)
- Integrations "media: connected" — computed from R2 env while actual path uses Supabase

5. External Integrations to Connect at End (B)

Resend (email) → Twilio (SMS) → WATI (WhatsApp) → Facebook (OAuth + leads pull) → Google Ads → Razorpay (online payments + webhook) → Inngest (activate event key — highest leverage: unlocks invite email, lead events, receipts, scheduled publish, outreach dispatch) → Gemini (AI tutor).

All have honest abstractions (notification.service.ts dispatch, crm/integrations status, events/inngest.ts gate, env.ts reserved keys). No fake success except the invite/settings toasts noted above.

6. Intentionally Deferred Features (C)

- Deals module (needs new Opportunity model + migration) — current /crm/deals is admission-funnel derived, honest not_implemented
- Voice AI (Vapi) call logs
- Error monitoring (Sentry/Datadog)
- LMS chat rooms (LmsChatRoom/LmsMessage) — tables exist, permission defined
- Public online payment (Razorpay checkout)
- Per-org Permission-table overrides (static matrix is authoritative)

7. Dead / Unused Code (D)

- Workflow / WorkflowRun models — zero src references
- FeatureFlag table — zero references (superseded by org.featureFlags JSON, itself only read by an unused gate)
- FallbackLead table — zero references
- LmsNotification + lms_notifications permission — zero service/UI
- LmsChatRoom/LmsMessage + lms_chat permission — no service/UI (deferred)
- isLmsEnabled (src/lib/lms/feature-flag.ts) — never imported
- assignCourseTeacherSchema — no route/UI

- LmsTimetableSlot.recurrenceRule — never written
- LmsCertificateStatus.REVOKED — no revoke path
- Nav API + ContentBlock API — fully built, zero UI
- api/v1/lms/dashboard/[studentId] — no UI consumer
- Alias redirect pages /crm/leads, /crm/pipeline, /crm/dashboard (intentional)
- Facebook OAuth callback page — plain redirect, no exchange
- /vapi, /errors, /notifications pages — orphan placeholders
- No service/repository is dead — all 22 services and 17 repositories are imported

8. P0 Fix List (Production Blockers)

1. **Courses "Go Live"** — courses/page.tsx:74 change PATCH → POST (verify with typecheck + live)
2. **Job application status** — reconcile OFFERED/HIRED with job.schema.ts + DB enum (SELECTED) + status-badge.tsx styles + transition map
3. **Settings fake-success** — wire applicationIntake/maintenanceMode into the public intake gate + middleware, or remove the controls; add /api/webhooks/* consumers or drop the fields
4. **Student self-grade vector** — remove STUDENT write,lms_assessments from permissions.ts (or gate me/assessments POST to quiz-derived sync only)
5. **Certificate print authorization** — add ownership/org check to /portal/certificates/[certNo]/print
6. **Certificate certNo generation** — move server-side with collision retry/catch (unhandled P2002 → 500)

9. P1 Fix List (Core Functionality Incomplete)

1. **Page editor UI** (sections/blocks/reorder/versions/rollback) + ContentBlock type seeding + real preview — server-side already complete
2. **Nav management UI** + nav public propagation
3. **Course manager completion** — curriculum editor, full edit dialog, versions UI, fix create-status strip, media picker
4. **Media completion** — folder create/rename/move/replace/bulk UI
5. **Notifications inbox** — Bell wiring, IN_APP delivery, inbox page; activate Inngest + Resend (or honest "not sent" surface on invite)
6. **Profile + password change** — wire "Profile Settings", build change-password route+UI, portal profile edit
7. **LMS assignments UI** — faculty list/submit/grade/submissions + fix mislabeled nav link
8. **LMS lifecycle routes** — attendance session/record edit+delete, certificate revoke/delete, course delete, unenroll
9. **Faculty students grouping**; build-or-remove dashboard-student UI
10. **Invite accept UI**; surface "email not sent" warnings
11. Minor: document rejection error type; student event name; scheduled-publish scheduler activation

10. Recommended Implementation Order

Sprint A — P0 correctness (do not ship without)

1. Courses "Go Live" method fix
2. Job application status enum reconciliation
3. Settings controls: wire intake/maintenance or remove; webhook honesty
4. Student self-grade fix (permissions)
5. Certificate print authorization + server-side certNo

Sprint B — P1 high-value UI completion (server-side already exists)

1. Page editor (sections/blocks/reorder/versions/rollback/preview) + ContentBlock seeding
2. Nav management UI + public nav
3. Course manager curriculum editor + full edit + versions
4. Media folder/replace/rename/move/bulk
5. LMS assignments UI + lifecycle routes (attendance edit/delete, cert revoke/delete, unenroll, course delete)
6. Notification inbox + Bell + activate Inngest (unlocks invite email, outreach dispatch, scheduled publish, receipts)

Sprint C — P2 hardening

1. Document error type, student event name, faculty grouping, unlock-rules evaluation, content-url UX, integrations status accuracy, activity-feed admin surface, honesty fixes (toasts/wording)

Sprint D — P3 + external

1. Deferred/dead cleanup (Workflow/FeatureFlag/FallbackLead/LmsNotification/chat — keep or drop), alias/placeholder pages
2. External providers (Resend → Twilio → WATI → Facebook → Google Ads → Razorpay → Gemini)

Explicitly out of scope per rules: auth changes, Elementor/Page Builder, Supabase storage re-architecture, CRM/CMS architecture changes, schema changes without verified necessity, dependency upgrades.